Unit 3

Introduction

The tunnel effect is a purely quantum effect since there is not a classical explanation. In 1954, the Nobel Prize for physics was awarded to Cockcroft and Walton for their initiation of nuclear physics twenty years previously. They made use of *quantum tunneling* to induce nuclear reactions in circumstances which classical physics deemed impossible. In 1973, the Nobel Prize was awarded to three scientists for their work in developing useful devices called *tunnel diodes* which depend for their operation on the same effect. These devices have opened up whole new areas of physics and technology. In 1986, another Nobel Prize was awarded to Binnig and Rohrer for their invention of the scanning tunneling electron microscope. All these prizes have been awarded for the exploitation of one quantum effect, carried out with great ingenuity and insight.

The behaviour of particles such as electrons, photons and nucleons is determined by their quantum mechanical characteristics, and we rely on the Born probability hypothesis to relate particle position to the amplitude squared of the associated wavefunction. Such particles may therefore be expected to exhibit wave-like characteristics.

When light waves encounter a boundary where the refractive index changes, there may be both a reflected and a transmitted wave. If the boundary is between a transparent and an opaque material, the wave will still penetrate into the opaque material for a distance roughly equal to the wavelength. It follows that light can be transmitted through a layer of a metal provided its thickness is less than the *penetration depth* of the wave. A good example of this is the layer of silver a few atoms thick on a glass slide forming a semi-reflecting mirror. In the context of quantum mechanics we might expect particles to show similar behaviour.

In the quantum mechanical case, the potential energy is roughly analogous to arefractive index. If the potential energy changes suddenly with position, then there is usually both a transmitted and a reflected quantum wave at the boundary, and the particles will be transmitted or reflected with calculable probability. If the increase in potential energy is greater than the particle kinetic energy, then the quantum wave will penetrate into the

classically forbidden region though its amplitude will rapidly decrease. In consequence, if the region of high potential energy is narrow enough, the wave will emerge with reduced amplitude on the other side and there is a probability that the associated particle will tunnel through. A narrow region of high potential energy is called a *potential barrier*. Quantum mechanics therefore predicts two effects totally alien to the classical mechanics based on Newton's laws: (i) particles may be reflected by any sudden change in potential energy, and (ii) particles can tunnel through narrow potential barriers. Experiment confirms both of these remarkable predictions.

Particle flux, reflection and transmission co-efficients

If there are N particles per unitlength and each one has speed u in the positive x-direction, then all the particles in a length $u\,\Delta t$ will pass a fixedpoint in time intervalΔt. The number of particles passing a fixed point per unit time is the particle flux (F)given by

$$F = \frac{Nu\Delta t}{\Delta t} = Nu \qquad (1)$$

In quantum mechanics, if wavefunction $\psi(x)$ represents a *single particle*, then $p(x)=|\psi(x)|^2$ gives the probability of finding the particle at the position x. The wavefunction can also represent *more than one particle*.

In particular, the wavefunction $\psi(x) = A\exp(ikx)$ can represent a set of particles, moving in the$x$-direction, all with the same momentum $\hbar k$. In this case, the position of any one particle is not specified (to do so violates the Heisenberg uncertainty principle), but the average number of particles per unit length is given by $|\psi(x)|^2= |A|^2$. In this context, 'average' means the average of a large number of observations made on particles with this wavefunction. The speed u is obtained from the momentum magnitude p according to the following relations

p = ħkand u = p/m

so that: u=ħk/m

Since $k > 0$, so is u.

where k is the wave-number.

The flux (F) is then given by

$$F = |A|^2 \frac{\hbar k}{m} \quad (2)$$

Note: Equation 2 can only be used when the wavefunction$\psi(x)$ is a momentum eigenfunction; the momentum is then the same for all the particles, and the average number of particles per unit length is a constant, independent of position.

Two coefficients (reflection coefficient and transmission coefficient) are defined which characterize the behavior of particles when they encounter a potential step.

The reflection coefficient (R) is defined as follows:

$$R = \frac{flux\ of\ reflected\ particles}{flux\ of\ incident\ particles} \quad (3)$$

The transmission coefficient (T) is defined as follows:

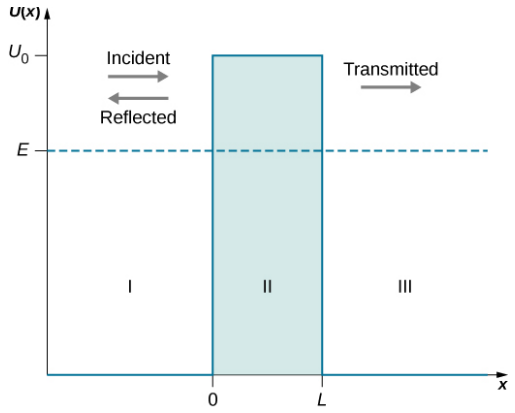
$$T = \frac{flux\ of\ transmitted\ particles}{flux\ of\ incident\ particles} \quad (4)$$

The reflection and transmission coefficients must sum to 1 as the number of particles is always conserved.

Reflection and transmission at a potential barrier

Suppose a uniform and time-independent beam of electrons or other quantum particles with energy E traveling along the positive x-axis encounters a potential barrier of width L and height U_0 described by

$$U(x) = \begin{cases} 0, when\ x < 0 \\ U_0, when\ 0 \leq x \leq L \\ 0, when\ x > L \end{cases} \quad (5)$$



When the height $U_0$of the barrier is infinite, the wave packet representing an incident quantum particle is unable to penetrate it, and the quantum particle bounces back from the barrier boundary, just like a classical particle. When the width L of the barrier is infinite and its height is finite, a part of the wave packet representing an incident quantum particle can filter through the barrier boundary and eventually perish after traveling some distance inside the barrier.

When both the width L and the height $U_0$are finite, a part of the quantum wave packet incident on one side of the barrier can penetrate the barrier boundary and continue its motion inside the barrier, where it is gradually attenuated on its way to the other side. A part of the incident quantum wave packet eventually emerges on the other side of the barrier in the form of the transmitted wave packet that tunneled through the barrier. How much of the incident wave can tunnel through a barrier depends on the barrier width and its height, and on the energy of the quantum particle incident on the barrier. This is the physics of tunneling.

The probability that an individual particle in the beam will tunnel through the potential barrier can be found by solving the boundary-value problem for the time-independent Schrödinger equation for a particle in the beam given by

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+U(x)\psi(x)=E\psi(x) \qquad (6)$$

where $-\infty < x < +\infty$

We assume that the given energy E of the incoming particle is smaller than the height $U_0$of the potential barrier, $E<U_0$, because this is an interesting physical case.

Denoting by $\psi_I(x)$ the solution in region I for x < 0, by $\psi_{II}(x)$ the solution in region II for 0 ≤ x < L, and by $\psi_{III}(x)$ the solution in region III for x > L, the stationary Schrödinger equation has the following forms in these three regions:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_I(x)}{dx^2} = E\psi_I(x),$$

$$\text{in region } I\text{: } -\infty < x < 0, \tag{7}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + U_0\psi_{II}(x) = E\psi_{II}(x)$$

$$\text{in region } II\text{: } 0 < x < L, \tag{8}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{III}(x)}{dx^2} = E\psi_{III}(x)$$

$$\text{in region } III\text{: } L < x < +\infty, \tag{9}$$

The general solutions of equations (7) and (9) are given by

$$\psi_I(x) = Ae^{+ikx} + Be^{-ikx} \tag{10}$$

$$\psi_{III}(x) = Fe^{+ikx} + Ge^{-ikx} \tag{11}$$

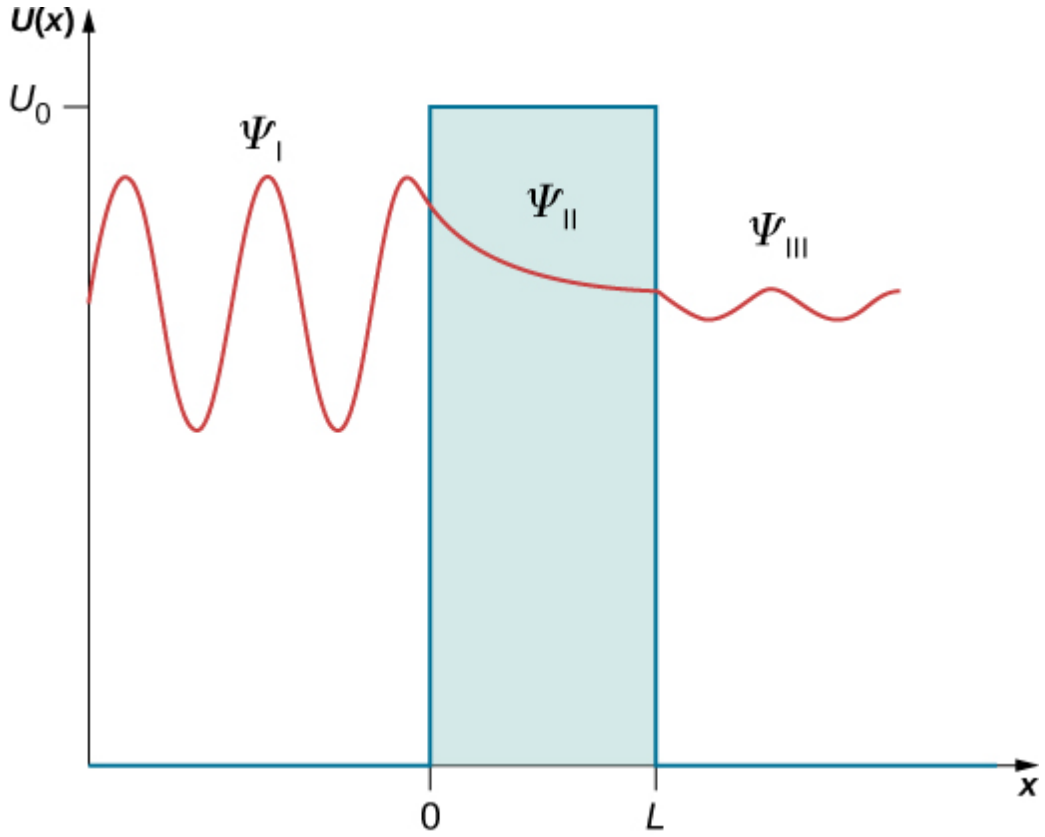
where the wave-number

$k = \sqrt{2mE}/\hbar$

the complex exponent denotes oscillations

$e^{\pm ikx} = \cos\ kx \pm i\ \sin\ kx.$

and the constants *A*, *B*, *F*, and *G* may be complex.



In region II, the terms in equation (8) can be rearranged to give

$$\frac{d^2\psi_{II}(x)}{dx^2} = \beta^2\psi_{II}(x) \tag{12}$$

where the real number β is given by

$$\beta^2 = \frac{2m}{\hbar^2}(U_0 - E) \tag{13}$$

and is positive for $U_0 > E$.

The general solution of eqn. (13) is not oscillatory (unlike in the other regions) and is in the form of exponentials that describe a gradual attenuation of $\psi_{II}(x)$

$$\psi_{II}(x) = Ce^{-\beta x} + De^{+\beta x} \tag{14}$$

In region I, there are two waves—one is incident (moving to the right) and one is reflected (moving to the left)—so the constants *A* and *B* in eqn. (10) are non-zero. In region III, there is only one wave (moving to the right), which is the transmitted wave, so the constant *G* must be zero.

The incident wave is represented as

$$\psi_{in}(x) = Ae^{+ikx} \tag{15}$$

reflected wave is denoted as

$$\psi_{ref}(x) = Be^{-ikx} \tag{16}$$

and the transmitted wave is given by

$$\psi_{tra}(x) = Fe^{+ikx} \tag{17}$$

The amplitude of the incident wave is

$$\begin{aligned} |\psi_{in}(x)|^2 &= \psi_{in}^*(x)\psi_{in}(x) \\ &= (Ae^{+ikx})^* Ae^{+ikx} \\ &= A^* e^{-ikx} Ae^{+ikx} \\ &= A^* A = |A|^2. \end{aligned} \tag{18}$$

Similarly, the amplitude of the reflected wave is

$$|\psi_{ref}(x)|^2 = |B|^2 \tag{19}$$

and the amplitude of the transmitted wave is

$$|\psi_{tra}(x)|^2 = |F|^2 \tag{20}$$

The transmission probability or tunneling probability *T(L,E)*, where *L*is the width of the barrier

and *E* is the total energy of the particle is given by

$$T(L,E) = \frac{|\psi_{tra}(x)|^2}{|\psi_{in}(x)|^2}$$
$$= \frac{|F|^2}{|A|^2}$$
$$= \left|\frac{F}{A}\right|^2 \qquad (21)$$

The unknown constant *A*, *B*, *C*, *D* and *F* are determined by the boundary conditions discussed below.

(i) The continuity condition at region boundaries (*I* and *II*; *II* and *III*) requires that:

$$\psi_I(0) = \psi_{II}(0) \qquad (22)$$

$$\psi_{II}(L) = \psi_{III}(L) \qquad (23)$$

(ii) The "smoothness" condition requires the first derivative of the solution be continuous at region boundaries (*I* and *II*; *II* and *III*)

$$\left.\frac{d\psi_I(x)}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}(x)}{dx}\right|_{x=0} \qquad (24)$$

$$\left.\frac{d\psi_{II}(x)}{dx}\right|_{x=L} = \left.\frac{d\psi_{III}(x)}{dx}\right|_{x=L} \qquad (25)$$

The substitution of equations (10) and (14) in eqn. (22) yields

$A + B = C + D$ (26)

Also, substitution of equations (11) and (14) in eqn. (23) yields

$$Ce^{-\beta L} + De^{+\beta L} = Fe^{+ikL} \qquad \text{[since, G = 0]} \tag{27}$$

Similarly, substitution of equations (10) and (14) in eqn. (24) yields

$$-ik(A - B) = \beta(D - C) \tag{28}$$

and the boundary condition (25) gives

$$\beta(De^{+\beta L} - Ce^{-\beta L}) = +ikFe^{+ikL} \tag{29}$$

We now have four equations for five unknown constants. However, on dividing eqns. (26-29) with A, we have only four unknown fractions: *B/A*, *C/A*, *D/A* and *F/A*. Three of these fractions (*B/A, C/A, D/A)* can be eliminated to give

$$\frac{F}{A} = \frac{e^{-ikL}}{cosh\,(\beta L) + i(\gamma/2)\,sinh\,(\beta L)} \tag{30}$$

where $\gamma \equiv \beta/k - k/\beta$

By substituting the definition of hyperbolic functions,

$$\cosh\, y = \frac{e^{y} + e^{-y}}{2} \quad ; \qquad \sinh\, y = \frac{e^{y} - e^{-y}}{2}$$

in equation (30), the tunneling probability *T(L,E)* is obtained as

$$T(L,E) = \left(\frac{F}{A}\right)^{*} \frac{F}{A} = \frac{e^{+ikL}}{\cosh\,(\beta L) - i(\gamma/2)\,\sinh\,(\beta L)} \cdot \frac{e^{-ikL}}{\cosh\,(\beta L) + i(\gamma/2)\,\sinh\,(\beta L)}$$

Or

$$T(L,E) = \frac{1}{\cosh^2(\beta L) + (\gamma/2)^2 \sinh^2(\beta L)} \tag{31}$$

where

$$\left(\frac{\gamma}{2}\right)^2 = \frac{1}{4}\left(\frac{1 - E/U_0}{E/U_0} + \frac{E/U_0}{1 - E/U_0} - 2\right)$$

For a wide and high barrier that transmits poorly, eqn. (31) can be approximated as

$$T(L,E) \approx 16\frac{E}{U_0}\left(1 - \frac{E}{U_0}\right) e^{-2\beta L} \tag{32}$$

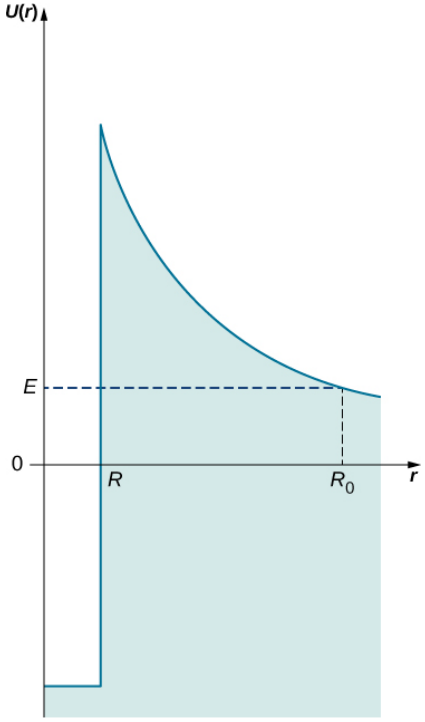
The tunneling effect very strongly depends on the width *L* of the potential barrier. In the laboratory, we can adjust both the potential height U_0 and the width to design nano-devices with desirable transmission coefficients.

Radioactive Decay

In 1928, Gamow identified quantum tunneling as the mechanism responsible for the radioactive decay of atomic nuclei. He observed that some isotopes of thorium, uranium, and bismuth disintegrate by emitting α-particles (which are doubly ionized helium atoms or, helium nuclei). In the process of emitting an α-particle, the original nucleus is transformed into a new nucleus that has two fewer neutrons and two fewer protons than the original nucleus. The α-particles emitted by one isotope have approximately the same kinetic energies. An examination of variations of these kinetic energies among isotopes of various elements shows that the lowest kinetic energy is about 4 MeV and highest is about 9 MeV, so these energies are of the same order of magnitude.

An inspection of the half-lives (a half-life is the time in which a radioactive sample loses half of its nuclei due to decay), indicates that the half-lives of different isotopes differ widely. For example, the half-life of polonium-214 is 160 μs and the half-life of uranium is 4.5 billion years. Gamow explained this variation by considering a ‘spherical-box’ model of the

nucleus, where α-particles can bounce back and forth between the walls as free particles. The confinement is provided by a strong nuclear potential at a spherical wall of the box. The thickness of this wall, however, is not infinite but finite, so in principle, a nuclear particle has a chance to escape this nuclear confinement. On the inside wall of the confining barrier is a high nuclear potential that keeps the α-particle in a small confinement. But when an α-particle gets out to the other side of this wall, it is subject to electrostatic Coulomb repulsion and moves away from the nucleus.



The width L of the potential barrier that separates an α-particle from the outside world depends on the particle's kinetic energy E. This width is the distance between the point marked by the nuclear radius R, and the point, R_0, where an α-particle emerges on the other side of the barrier,

$$L = R_0 - R \tag{33}$$

At the distance the kinetic energy of the α-particle must be equal to electrostatic Coulomb repulsion energy.

$$E = (4\pi\epsilon_0)^{-1} Ze^2/R_0 \tag{34}$$

Substituting the value of R_0 obtained from eqn (34) into (33) gives,

$$L = \frac{e^2}{4\pi\epsilon_0}\frac{Z}{E} - R. \tag{35}$$

Hence, higher the energy of α-particle, the narrower the width of the barrier that it is to tunnel through. Also, the width of the potential barrier is the most important parameter in tunneling probability. Thus, highly energetic α-particles have a good chance to escape the nucleus, and, for such nuclei, the nuclear disintegration half-life is short. However, this process is highly nonlinear, meaning a small increase in the α-particle energy can exponentially increase the tunneling probability (given by eqn. 32). This explains why the half-life of polonium that emits 8-MeV α-particles is only hundreds of milliseconds and the

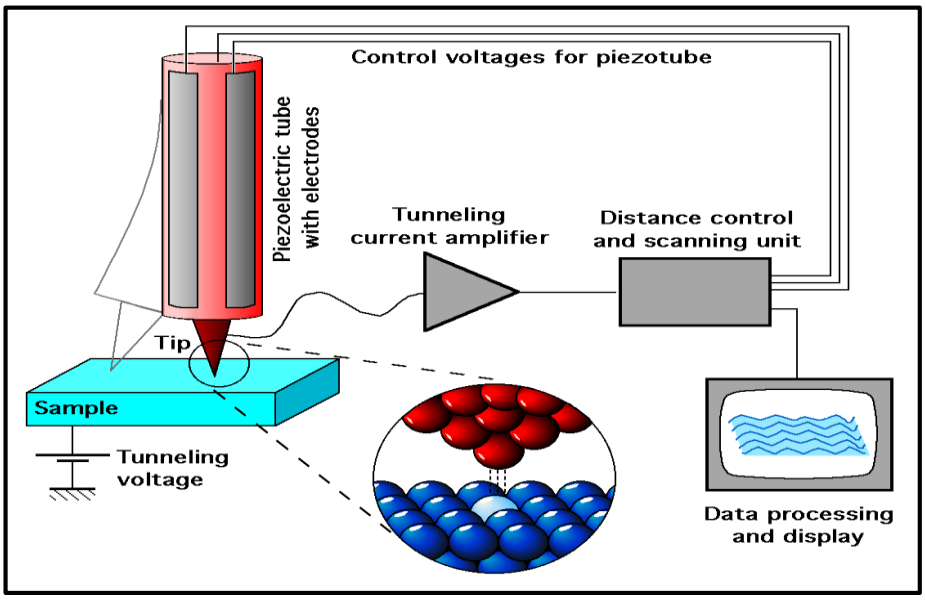
half-life of uranium that emits 4-MeV α-particles is billions of years.

Scanning tunneling microscope

Field emission is a process of emitting electrons from conducting surfaces due to a strong external electric field that is applied in the direction normal to the surface. An applied external electric field causes the electrons in a conductor to move to its surface and stay there as long as the present external field is not excessively strong. This gives rise to a constant electric potential throughout the inside of the conductor, including its surface. Hence, an electron inside the conductor has a constant potential energy which serves as the potential energy barrier. When the external electric field is strong, conduction electrons at the surface may get detached from it and accelerate along electric field lines in a direction antiparallel to the external field, away from the surface. In short, conduction electrons may escape from the surface. The field emission can be understood as the quantum tunneling of conduction electrons through the potential barrier at the conductor's surface.

This quantum tunneling phenomenon at metallic surfaces, is the physical principle behind the operation of the scanning tunneling microscope (STM), invented in 1981 by Gerd Binnig and Heinrich Rohrer. The STM device consists of a scanning tip (a needle, usually made of tungsten, platinum-iridium, or gold); a piezoelectric device that controls the tip's elevation in a typical range of 0.4 to 0.7 nm above the surface to be scanned; a device that controls the motion of the tip along the surface; and a computer to display images. While the sample is kept at a suitable voltage bias, the scanning tip moves along the surface and the tunneling-electron current between the tip and the surface is registered at each position.

The amount of the current depends on the probability of electron tunneling from the surface to the tip, which, in turn, depends on the elevation of the tip above the surface. Hence, at each tip position, the distance from the tip to the surface is measured by measuring how many electrons tunnel out from the surface to the tip. This method can give an unprecedented resolution of about 0.001 nm, which is about 1% of the average diameter of an atom. When the strength of an external electric field is constant, the tunneling-electron current has different values at different elevations of the tip above the surface.



Quantum Harmonic Oscillator

Harmonic motion occurs when a system of some kind vibrates about an equilibrium configuration such as simple pendulum, block on a spring, electromagnetic waves, vibrating molecules etc. The condition of harmonic motion is the presence of a restoring force which returns the system to its equilibrium configuration when it is disturbed. The harmonic oscillator is a fundamental problem of classical mechanics as well as quantum mechanics. Classical mechanics can be effectively employed to study macroscopic oscillations where the de Broglie wavelength of the particle is much less than the amplitude of oscillation. On the other hand, if the de Broglie wavelength of the particle is comparable to the amplitude of oscillation such as vibrations of molecules, then the motion of the particle can be described by quantum mechanics.

The Schrödinger's equation for a harmonic oscillator having potential $U = \frac{1}{2}kx^2$ is given by

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{1}{2}\,kx^2\right)\psi = 0 \qquad (1)$$

where E is the energy of the particle and k is the force constant which is related to the frequency of oscillation ν and mass m of the particle by the relation

$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad (2)$$

Eq. (1) can be simplified by introducing the dimensionless quantities

$$y = \left(\frac{1}{\hbar}\sqrt{km}\right)^{1/2} x \text{ and } \alpha = \frac{2E}{h\nu} \qquad (3)$$

The purpose of introducing these quantities in Eq. (1) is to change the units in x and E to dimensionless units.

Schrödinger's equation in terms of y and α is given by

$$\frac{d^2\psi}{dy^2} + (\alpha - y^2)\psi = 0 \qquad (4)$$

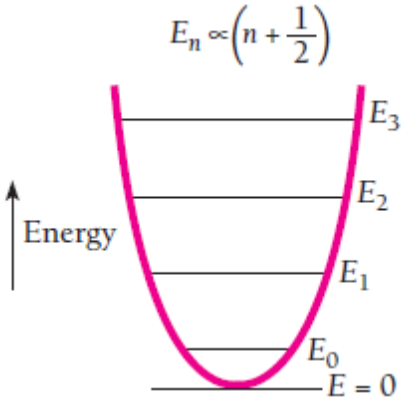
The acceptable solutions to this equation are limited by the condition that $\psi \to 0 \; as \; y \to \infty$. This condition is fulfilled when

α = 2n + 1, where n = 0, 1, 2, 3, ….. (5)

Substituting the value of α from Eq. (3) in Eq. (5), the energy levels of a harmonic oscillator with frequency of oscillation ν is given by

$$E_n = \left(n + \frac{1}{2}\right) h\nu, \quad \text{n = 0, 1, 2, 3, ….} \qquad (6)$$

It is observed that the energy of a harmonic oscillator is quantized and the energy levels are all equally spaced.



The energy corresponding to the ground state i.e, n = 0 is

$$E_0 = \frac{1}{2} h\nu$$

The lowest value of energy of a harmonic oscillator is called zero-point energy because a harmonic oscillator in equilibrium with its surroundings will approach this value and not E = 0 as temperature approaches 0K.

The general formula for the nth normalized wave function of a harmonic oscillator in terms of the Hermite polynomial $H_n(y)$is given by

$$\psi_n = \left(\frac{2m\nu}{\hbar}\right)^{1/4} (2^n n!)^{-1/2} H_n(y) e^{-y^2/2} \qquad (7)$$

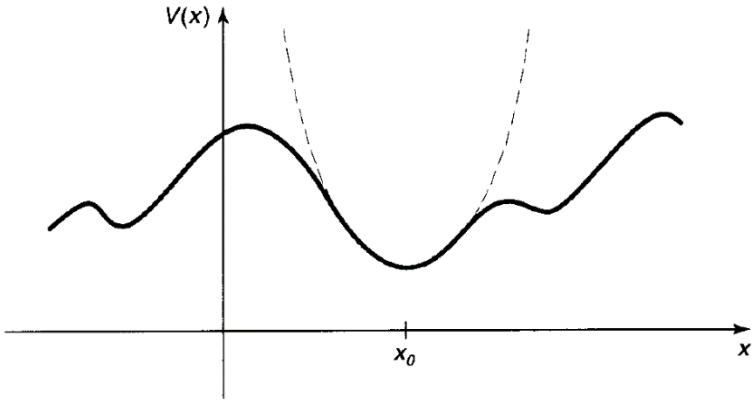
The lowest-order Hermite polynomials are given below:

$$H_0(y) = 1; \; H_1(y) = 2y; \; H_2(y) = 4y^2 - 2; \; H_3(y) = 8y^3 - 12y$$

Significance of harmonic oscillator in quantum mechanics

The significance of harmonic oscillator in quantum mechanics is that any particle that is in its position of a stable equilibrium will execute simple harmonic motion (SHM) if displaced by a small amount. For example, simple pendulum or atom of hydrogen in hydrogen chloride (HCl).

Intermolecular potential energy vs. intermolecular distance for a diatomic molecular distance is shown in the figure alongside.



Parabolic approximation (dashed curve) to an arbitrary potential, in the neighborhood of a local minimum.

Example 7

Find the expectation value <x> for the first two states of harmonic oscillator.

Given: $\psi_0 = \left(\frac{2mv}{\hbar}\right)^{1/4} e^{-y^2/2}$ $\psi_1 = \left(\frac{2mv}{\hbar}\right)^{1/4} \left(\frac{1}{2}\right)^{1/2} (2y) e^{-y^2/2}$ $y = \sqrt{\frac{2\pi mv}{\hbar}}\, x$

Solution:

$$\langle x \rangle = \int_{-\infty}^{\infty} x|\psi|^2 dx$$

We will initially begin with y and then replace it by x.

n = 0; $\int_{-\infty}^{\infty} y|\psi_0|^2 dy = \int_{-\infty}^{\infty} y e^{-y^2} dy = 0$

n = 1; $\int_{-\infty}^{\infty} y|\psi_1|^2 dy = \int_{-\infty}^{\infty} y^3 e^{-y^2} dy = 0$

<x> = 0

Hydrogen Atom

The constituents of a hydrogen atom are a proton and an electron. For the sake of simplicity, it is assumed that the proton is stationary and the electron is revolving around it.

The Schrödinger's equation for the electron in three dimensions i.e., hydrogen atom is

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2m}{\hbar^2}(E - U)\psi = 0 \qquad (1)$$

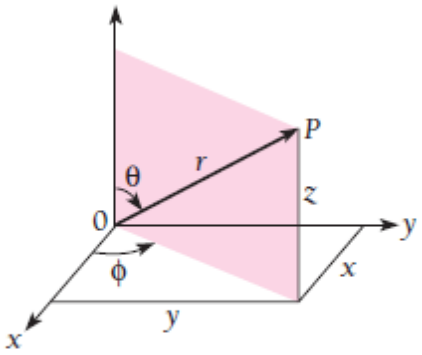
Here, U represents the electric potential energy between the proton and electron separated by a distance r and is given by

$$U = -\frac{e^2}{4\pi\varepsilon_0 r} \qquad (2)$$

Substituting Eq. (2) in Eq. (1), we get

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2m}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right)\psi = 0 \qquad (3)$$

It is difficult to solve Eq. (3) in Cartesian co-ordinates. However, if the Cartesian co-ordinates (x, y, z) are transformed to spherical polar co-ordinates (r, θ, ϕ), then Eq. (3) can be easily solved. Here, r is the length of the radius vector from the origin O to the point P.



θ represents the angle between the radius vector and +z axis. It is termed as the zenith angle.

ϕrepresents the angle between the projection of the radius vector in the xy plane and +x axis. It is termed as the azimuthal angle.

The Cartesian co-ordinates are related to spherical co-ordinates by the following relations.

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

The Schrödinger's equation in spherical polar co-ordinates is given by

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}+\frac{2m}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)\psi=0 \quad (4)$$

The above equation can be separated easily into three independent equations, each involving only one variable. After carrying out appropriate substitutions and separating the variables, the following three total differential equations are obtained

Azimuthal wave equation

$$\frac{d^2\Phi}{d\phi^2}+m_l^2\Phi=0 \quad (5)$$

Polar wave equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left[l(l+1)-\frac{m_l^2}{sin^2\theta}\right]\Theta=0 \quad (6)$$

Radial wave equation

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left[\frac{2m}{\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_0 r}+E\right)-\frac{l(l+1)}{r^2}\right]R=0 \quad (7)$$

The solution of the azimuthal wave equation (5) is given by

$$\Phi(\phi)=Ae^{im_l\phi} \quad (8)$$

The condition that the wave function Φ must have a single value at a given point in space. Now, since ϕ and ϕ + 2π represent a single point in space $Ae^{im_l\phi}=e^{im_l(\phi+2\pi)}$

This is possible only if m_l is 0 or a positive or negative integer.

i.e., $m_l=0,\pm1,\pm2,\pm3,$ …….

The constant m_l is known as the magnetic quantum number.

The polar wave equation (6) has a complex solution in terms of associated Legendre functions provided that the constant $l$in Eq. (6) is an integer equal to or greater than the absolute value of the magnetic quantum number i.e., $|m_l|$. This implies

$m_l=0,\pm1,\pm2,\pm3,$ ……., $\pm l$.

The constant l is termed as the orbital quantum number.

The radial wave equation (7) has a complex solution in terms of associated Laguerre polynomials under the following conditions.

(i) The energy E in eq. (7) have one of the negative values E_n(signifying that the electron is bound to the atom) given by

$$E_n = -\frac{me^4}{32\,\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n^2}\right) = \frac{E_1}{n^2} \quad n = 1, 2, 3, \ldots. \qquad (9)$$

(ii) The constant n in Eq. (9), known as the principal quantum number must be equal to or greater than l + 1.This condition means that

$$l = 0, 1, 2, 3, \ldots\ldots\ldots, (n-1)$$

Hence, the three quantum numbers needed to describe the three-dimensional motion of the electron in the hydrogen atom are n, l and m_l and they must satisfy the following conditions.

(a) Principal quantum number, $n \geq 1; n = 1, 2, 3, \ldots.$
(b) Orbital quantum number, $l \leq (n-1); l = 0, 1, 2, 3, \ldots\ldots\ldots\ldots\ldots\ldots, (n-1)$
(c) Magnetic quantum number, $|m_l| \leq l;\ m_l = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots, \pm l$

The wave function for the ground state of hydrogen atom is

$$\psi_1 = \left(\frac{1}{\sqrt{\pi}a_0^{3/2}}\right)e^{-r/a_0} \qquad (10)$$

where a_0 is the radius of the innermost orbit of the hydrogen atom also termed as first Bohr radius.

The probability P(r)dr of finding an electron at any radial distance within a spherically symmetric shell of thickness dr and volume $4\pi r^2 dr$ is given by

$$P\,(r)dr = |\psi_n|^2 4\pi r^2 dr$$

The probability of finding the electron in the ground state of the hydrogen atom is given by

$$P\,(r)dr = |\psi_1|^2 4\pi r^2 dr = \psi_1^*\psi_1 4\pi r^2 dr = \left[\frac{1}{\pi a_0^3}e^{-2r/a_0}\right]4\pi r^2 dr$$